

Nathaniel del Rosario

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EDUCATION

University of California, Berkeley

Berkeley, CA

M.S. Data Science

2026

- Artificial Intelligence, ML Systems, Data Engineering, Reinforcement Learning, Experimentation & Causal Inference

University of California, San Diego

La Jolla, CA

B.S. Data Science

March 2025

- Graduate (cross enrollment): Computer Vision, Recommender Systems, Deep Learning
- Principles & Techniques of Data Science, Statistics, Relational Databases, Operating Systems, Cloud Computing, Scalable ML

SKILLS

Python, Pandas, NumPy, PyTorch, Sci-Kit Learn, Langchain, ChromaDB, HuggingFace, OpenCV, Dash/Plotly, Dask, Streamlit AWS, Google Cloud, Docker, Kubernetes, Airflow, SSH, Snowflake, Databricks, BigQuery, PostgreSQL, Github, Jira, ArcGIS

EXPERIENCE

American Express

June 2026 - August 2026

Data Science Intern II - Global Decision Science, AI Lab & Governance

New York

- Designed & implemented novel Multi-Agent classification pipeline reducing manual monthly review volume (164000) by 50%
- Achieved recall and precision lifts of 7% and 6% through different agent architectures & stratified false positive rate analysis.
- Parallelized large workloads to reduce batch inference time by up to 20x

CVS Health

May 2025 - August 2025

Data Science Intern - Analytics & Behavioral Change

New York

- Democratized knowledge base of 39 million members & 77,000 unique diagnoses by implementing an Agentic-RAG Text-to-SQL tool leveraging BigQuery, Langchain, Vertex AI, ChromaDB, Docker, Kubernetes, Datadog Eppo
- Integrated MVP into existing API, optimizing performance between 82% to 93% across selected metrics through grid search
- Investigated causality of 330 million medical equipment claims, identifying potential cost-saving opportunities of up to \$21 million per successful rollout

Hacioglu Data Science Institute

September 2024 - March 2025

Undergraduate Research Assistant - Zhiting Hu's Group

La Jolla

- Ran compute v.s. scaling experiments utilizing Monte Carlo Tree Search improving by 10% on OSWorld Benchmark
- Integrated OSWorld support into the LLM-Reasoners Python library (2k Github Stars) by implementing a Self-Evaluation Reward Function and custom Search Algorithm Configurations based on current state and action trajectory

University of California, San Diego

September 2023 - March 2025

Instructional Assistant - Halicioglu Data Science Institute

La Jolla

- Beta Testing assignment and exam questions, hosting Office Hours for a data science course of over 500 students
- Updated deployment of course website using github pages & Docker supervised by under Suraj and Tiefenbruck
- Grading and hosting Office Hours for upper division data science course of over 700 students under Shannon Ellis, Sam Lau

Bio-Rad

June 2024 - August 2024

Data Science Intern - Clinical Diagnostics Group

Pleasanton

- Leveraged AWS RDS, Docker, PostgreSQL to deploy database, reduced storage usage by 35% through schema optimization
- Built ETL pipeline w/ Pandas, RestAPI to ensure 100% data integrity & improved consistency from 91% to 99.9%
- Utilized AWS EC2 to deploy web-app & unit tests utilizing Dask to achieve 5.1x / 80.1% speedup on data validation

San Diego Supercomputer Center

June 2023 - September 2023

Machine Learning Engineer Intern

Remote

- Designed Content-Based Filtering Recommender System utilizing Cosine and Jaccard similarity for baseline output
- Deployed Recommender System on AWS EC2, Lambda, achieving a design that scaled to process 200,000+ points
- Trained an RL agent using Stable Baselines and Q-Learning to improve recommendation quality after 100 iterations
- Utilized AWS S3, PostgreSQL for database queries & vectorized code to achieve 1.7x runtime speedup in feature engineering

Deloitte

February 2023 - June 2023

Data Science Fellow

Remote

- Cleaned data w/ 3000+ features, 1 billion observations using Dask, vectorized Pandas to decrease cleaning runtime by 20%
- Leveraged XGBoost, Lasso to identify 850 significant features, predict drug use in young adults with 81% accuracy
- Tuned Hyperparameters, class weighting to improve F1 score from .35 to .70 and identify 10 highest risk demographics

Chan Zuckerberg Biohub

June 2022 - January 2023

Data Science Intern - Data Science & Computational Biology Platform

San Francisco

- Built 9 interactive visualizations of CRISPR screen comparisons between 20000 features using Pandas/Dash/Plotly
- Improved data processing of a Nextflow data pipeline (16,000,000 data points) to minimize runtime by 10%
- Designed algorithms to compare across 30+ virus screens to yield insights in virus-host interactions using vectorized code